

Self-supervised Learning with Synthetic Negatives

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Introduction

Reconstruction-based anomaly detection provides a natural unsupervised baseline for jet anomaly detection, but in practice we found it difficult to make robust. Its performance relies heavily on fine-tuned hyperparameters and decoder failure.

We move toward **representation-based anomaly detection**, where the goal is to learn a jet event representation that encodes semantic information meaningfully.

We build on the **Particle Transformer (ParT)** and explore a **LeJEPa-style** self-supervised objective. We further incorporate synthetic corrupted data and a **Triplet loss** to explicitly encourage separation of the corrupted events in representation.

Our first attempt achieves stable anomaly detection performance, reaching a highest **AUC of 0.642**, which is already close to Transformer-AE's **previous highest of 0.661**. However, the performance **does not degrade during training**, and it can be scaled to substantially larger model sizes.

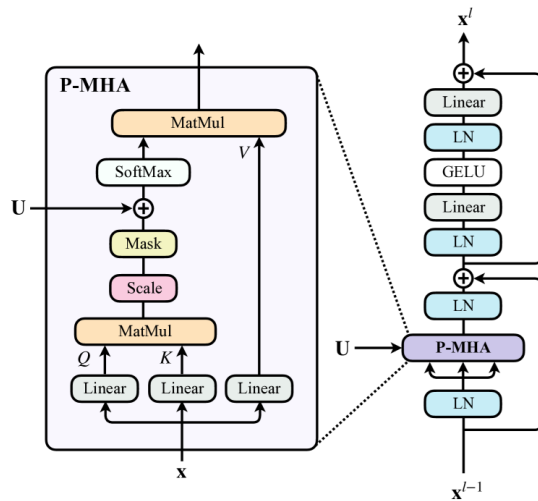
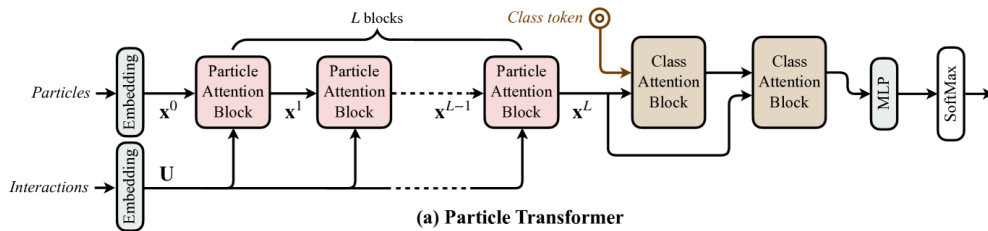
Architecture

ParT

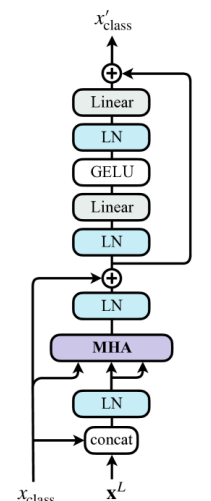
We adopt the **Particle Transformer (ParT)** as the backbone architecture

ParT is a Transformer-based **jet-tagging architecture** that treats a jet as a permutation-invariant set of particles and injects **pairwise particle-interaction features** as attention biases, allowing the model to learn both per-particle and jet-level global representation with strong inter-particle bias.

ParT uses a supervised classification target, where the final CLS hidden state is projected with a multi-class classification head. We modify the head to output **a representation vector** instead.



(b) Particle Attention Block



(c) Class Attention Block

Architecture

ParT

We use the following pairwise features for attention bias.

The 4 features are passed to a **3-layer MLP** to obtain attention biases for each head. The biases have shape **($B * \text{num_heads}$, N , N)**, where B is batch size, and N is number of PFCands.

Every self-attention layer receives the same set of biases, which is consistent with the standard ParT design.

```
1.  $\log(1 + \Delta R)$   
2.  $\log(1 + \min(\text{pt}_i, \text{pt}_j) * \Delta R)$   
3.  $\log(1 + \min(\text{pt}_i, \text{pt}_j) / (\text{pt}_i + \text{pt}_j))$   
4.  $\log(1 + m_{ij}^2)$ 
```

where m_{ij} is the invariant mass of the pair.

Optimization Target

LeJEPA (Lean Joint-Embedding Predictive Architecture) learns jet representations by making the model predict the representation of **augmented views** from another

Regularization such as **SIGReg** used to prevent representation collapse by forcing an **isotropic Gaussian** representation distribution.

LeJEPA argues that an isotropic Gaussian is the optimal distribution, as it minimizes worst-case downstream prediction risk.

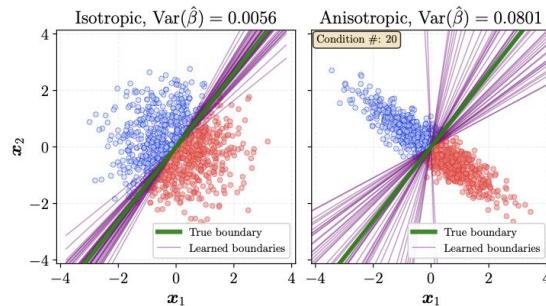


Figure 3. Illustration of lemma 2 showcasing how anisotropic (right) embeddings lead to higher variance estimator compared to isotropic embeddings (left). We sample 100 training points for the 2-class classification task and fit a logistic regression—repeating the process over numerous training set sample. Each sampling results in a decision boundary (purple).

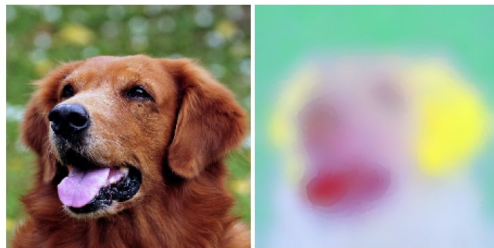


Figure 14. LeJEPA learns rich semantic representations through self-supervised learning. PCA visualization of last-layer features from LeJEPA (ViT-Large, 100 epochs on ImageNet-1K). For each image, features are independently projected to RGB using the first 3 principal components. Without any supervision, LeJEPA spontaneously develops semantically meaningful representations: notice how warm colors (red/-magenta/ pink) consistently capture foreground objects (parrot bodies, dog face), while cool colors (cyan/ green/ yellow) represent backgrounds and foliage. This emergent object-background separation and perceptual grouping discovered the visual structure of the world purely from unlabeled data.

<https://arxiv.org/pdf/2511.08544>

Optimization Target

LeJEPA consists of two components: **Invariant Loss and SIGReg**

Invariant loss claims the semantic meaning of data should be consistent or close under augmentation

1. Generate **global views** and **local views** of original data (mild and large augmentation respectively)
2. Forward pass all views through the model to get representations
3. Define anchor representation as mean of global representations
4. Invariant loss is defined as the mean MSE between each view and the anchor

SIGReg calculates the **distance** from output representation distribution to isotropic Gaussian

1. Randomly projecting high-dimensional representations onto many 1D directions
2. Apply an Epps–Pulley characteristic-function test on each projection

SIGReg gives high loss when the output collapses, low loss when it is close to isotropic Gaussian.

Data Augmentation

Generate Views

Randomly distorting PFCand features might violate physics rules. We adopt a conservative approach which only randomly drops PFCands.

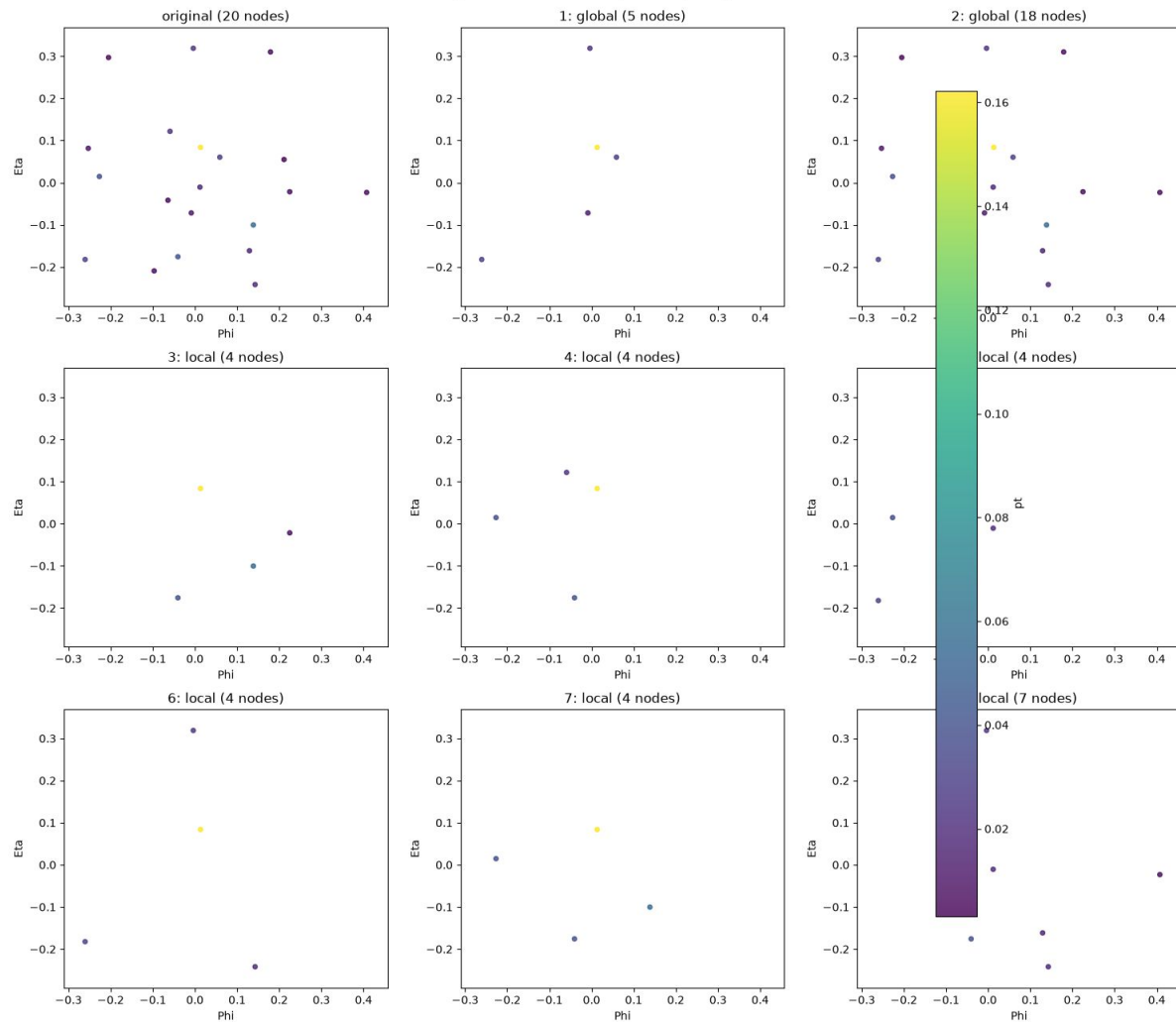
This is a **cumulative-pT-controlled multi-view augmentation**. For each jet and each view, we randomly sample a target dropped pT fraction, then generate a low-pT-biased random ordering of particles. Particles are masked out following this order until the cumulative dropped pT reaches the sampled target, while keeping at least a minimum number of valid particles.

1. The ranking score of each PFCand i is given by $-\log(u_i) * pt_i$, where $u_i \sim \text{Uniform}(0, 1)$.
2. For global views, we drop 0~50% pt in an event.
3. For local views, we drop 50~95% pt in an event.
4. For all views, we keep a minimum of 4 PFCands.

This plot shows an
example of original jet,
its 2 global views,
and 6 local views.

x-axis = phi
y-axis = eta
color = pt

All data are scaled



Data Augmentation

Synthetic Negatives

LeJEPa provides a "contraction" force to structure and tighten the representation space. However, a "repulsion" force is also needed so anomalous samples' representations are separated from normal representations.

To do this, we incorporate a Triplet Loss which enforces the distance between anchor and negative samples should be at least some value further than that between anchor and positive samples.

- Anchor = mean representation of global views
- Positives = representations of global views
- Negatives = representations of synthetic, corrupted jets

Data Augmentation

Synthetic Negatives

Identity_shuffle:

With in each event, shuffle the identity block jointly: charge, mass, pdgId one-hot

pt_resample:

Within each event, resample pt values with replacement from that event's valid pt/log_pt set (keeping pt and scaled log_pt together). Then renormalize the pt sum to the original event pt sum and match the original event-level log_pt mean/std.

eta_phi_shuffle:

Within each event, shuffle the eta-phi pair jointly across nodes.

batch_mix:

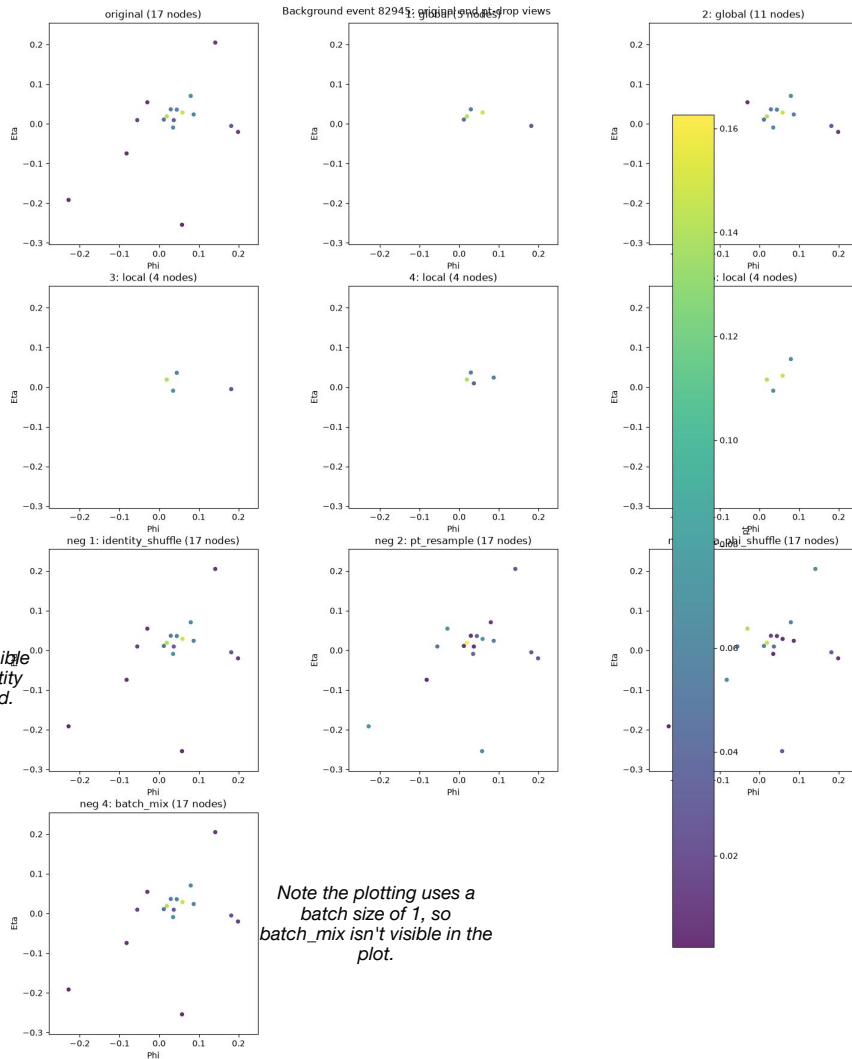
Build a fake jet by concatenating valid nodes from the anchor event and another donor event in the same batch. The resulting pt sum is renormalized to the original anchor event's pt sum, and log_pt is matched to the original anchor event's mean/std.

This plot shows an example of original jet, its 2 global views, 3 local views, and 4 negative samples.

x-axis = phi
y-axis = eta
color = pt

All data are scaled

Identity shuffle isn't visible in the plot, since identity block is not visualized.



Note the plotting uses a batch size of 1, so batch_mix isn't visible in the plot.

Data Set

We use our old QCD vs Wjet data set with pt limited between 200 and 400.

We use 99,979 background train QCD events, 24,995 background val QCD events, and 4,064 signal Wjet events.

PFCand features include:

- Eta, phi, pt, d0/d0Err, dz/dzErr, charge, mass, log_pt, pdgId (one-hot)

Hyper-parameters

We generally keep the same model hyper-parameters with the original ParT implementation:

- 8 Transformer Encoder Layers
- 2 CLS Pooling Layers
- 128 Hidden Size
- 128 Representation Size
- 2 Global Views + 6 Local Views (LeJEPA only)
- 2 Global Views + 3 Local Views + 4 Synthetic Negatives (LeJEPA + Triplet)
- 50 Epochs
- 128 Batch Size
- $5e-4$ Learning rate with cosine annealing (LeJEPA default)
- $5e-2$ Weight decay (LeJEPA default)

Anomaly Score

We no longer use reconstruction MSE as the anomaly score. Instead, we use **Mahalanobis distance** between an event and the training representation distribution as the anomaly score.

Mahalanobis distance measures how far a point is from a reference distribution while accounting for the covariance structure of the reference, so deviations along high-variance directions are penalized less than deviations along low-variance directions.

$$d_M(\vec{x}, Q) = \sqrt{(\vec{x} - \vec{\mu})^\top \Sigma^{-1} (\vec{x} - \vec{\mu})}.$$

We expect anomaly data to have representations “far away” from the training data.

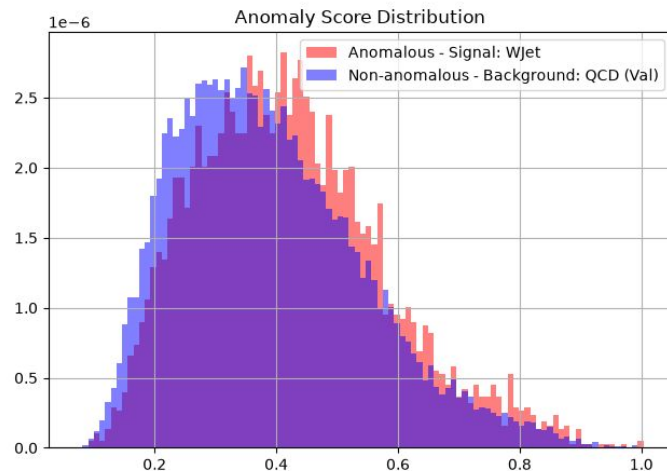
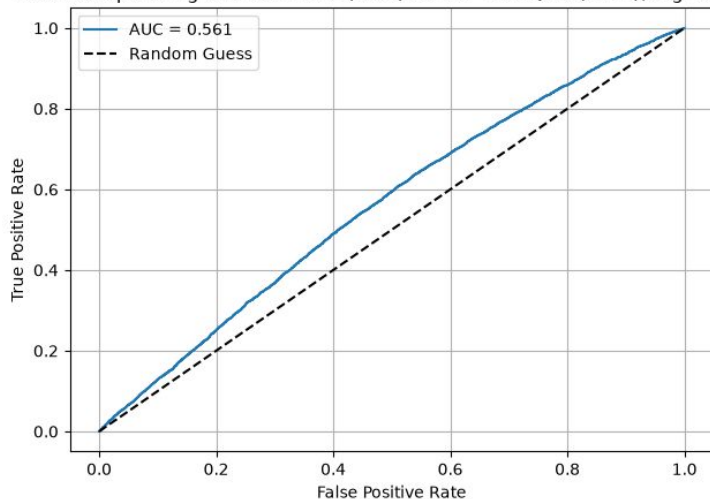
Results

LeJEPA Only

The first attempt with **just LeJEPA loss** achieves an **AUC of 0.561**. The model is trained to match views only, so there isn't an explicit "repulsion" force to encourage it to separate representations of "valid" data and "bad" data.

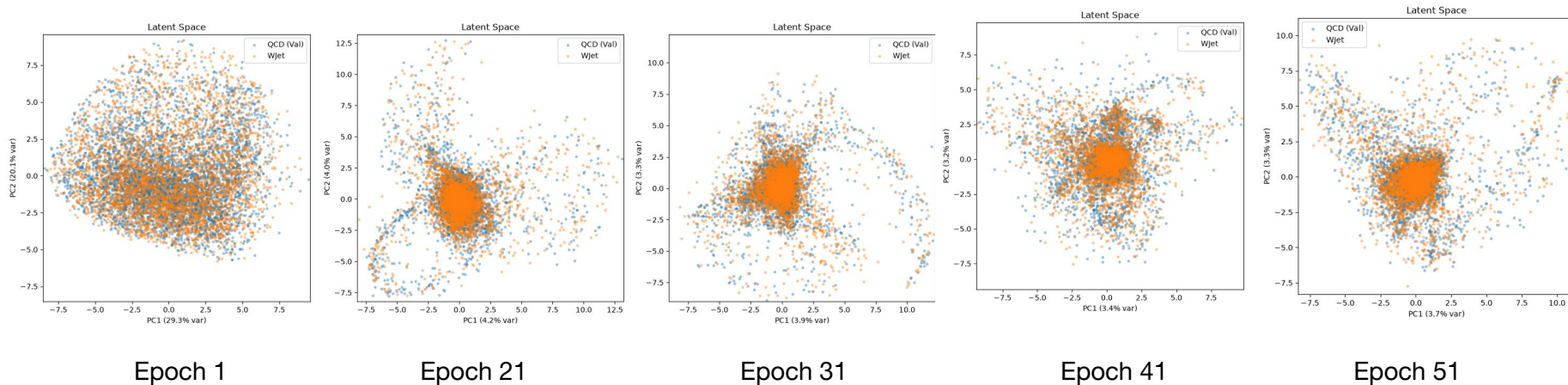
We did not observe significant difference between train-vs-signal ROC and val-vs-signal ROC. Suggesting little overfit at this scale.

Receiver Operating Characteristic (ROC) Curve - BG: QCD (Train), Signal: γ



Results

LeJEPA Only



In PCA of representation space, we barely observe any distributional difference between QCD (val) and Wjet. But this is expected, because we are only able to show two principle components, but there are 128 representation dimensions.

Results

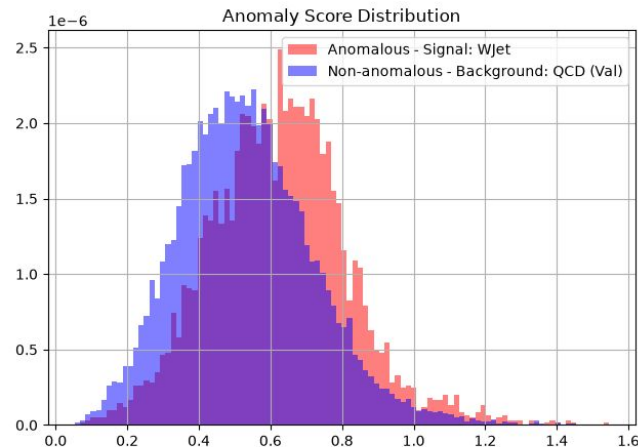
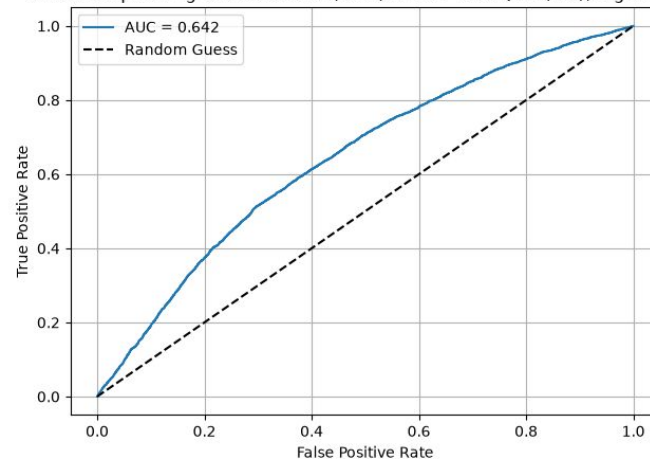
LeJEPA + Triplet Loss

To introduce a “repulsion force” into the model, we extended LeJEPA to include explicit negative contrast, to encourage a more defined latent space such that **anomaly-looking jets** are far away from the valid ones.

We observe a **significant boost in AUC** which reaches **0.642**.

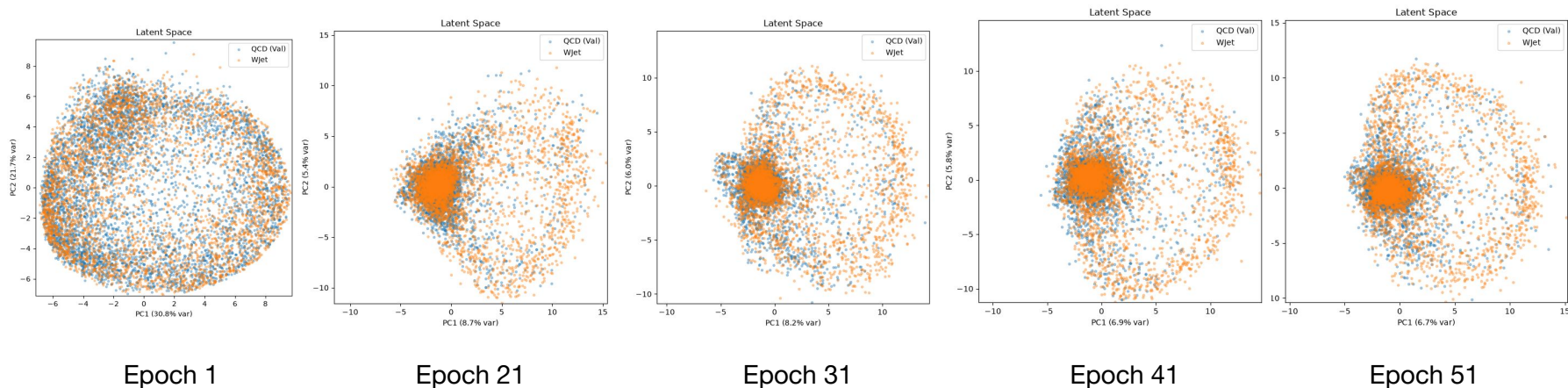
$$L_{\text{triplet}} = \text{ReLU}(\text{MSE}(\text{anchor}, \text{positive}) - (\text{MSE}(\text{anchor}, \text{negative}) - \text{margin}))$$

Receiver Operating Characteristic (ROC) Curve - BG: QCD (Val), Signal: W

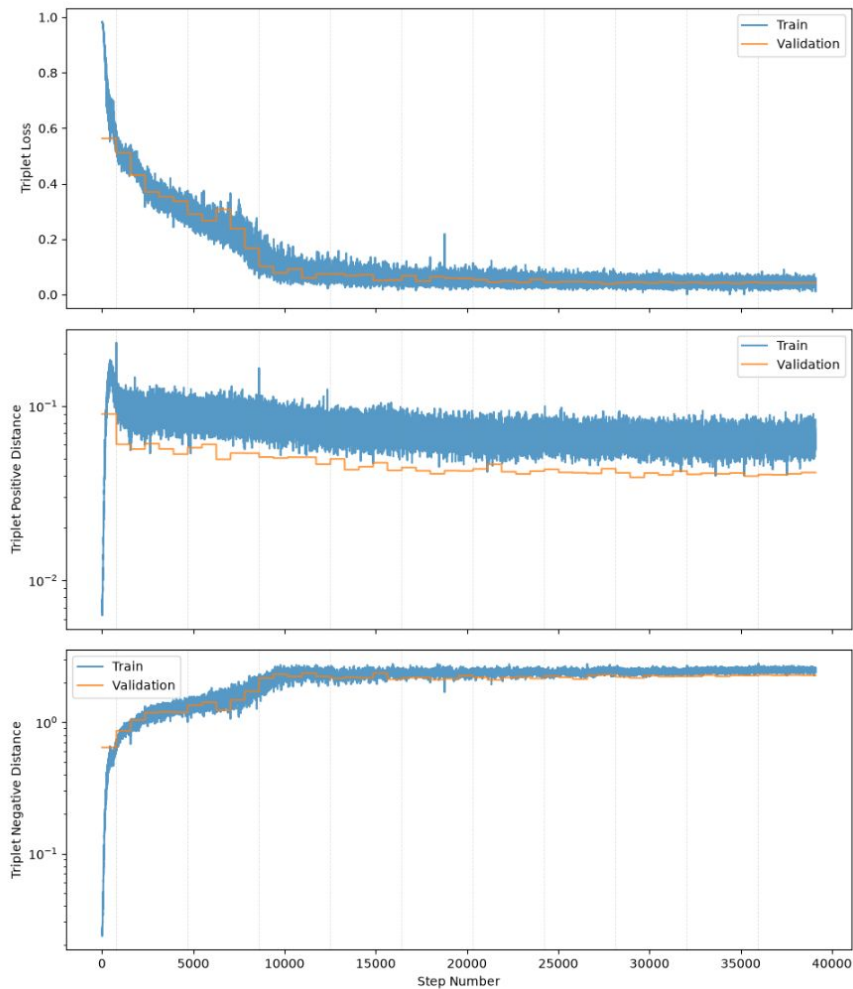
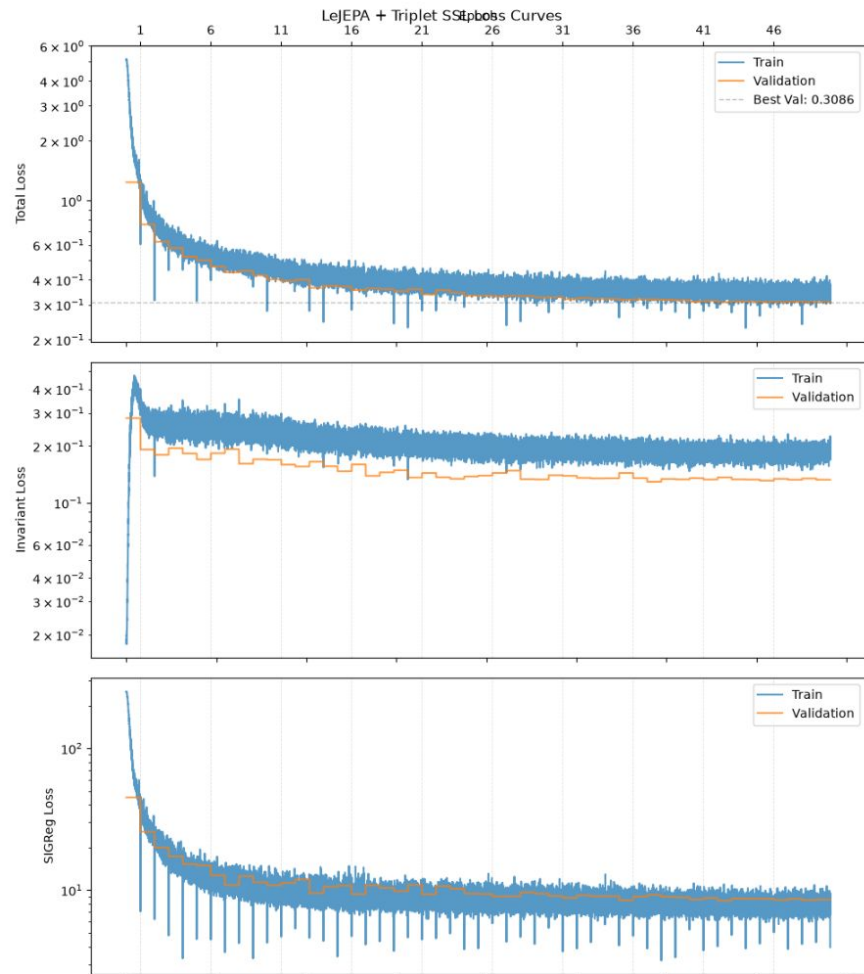


Results

LeJEPA + Triplet Loss



The PCA of representation space is different from pure LeJEPA model, but show a more stable distribution over epochs.



Next Steps

- Increasing train data amount and diversity (JetClass dataset, GRID certificate). Currently, the model is undertrained.
- Mixed training with known anomalies or different jet types
- Exploring different hyper-parameter settings (PFCand features, view counts, pairwise features for bias)
- Further exploring data augmentation techniques
- Experiment for longer epochs and scale the model up